

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x) = \log_y(x^{4y}) = 10$. Find xy .

Reasoning + Answer:

<tool_call> reasoning process here </tool_call>

We are given that there exist real numbers x and y , both greater than 1, such that:

$$\log_x(y^x) = 10 \quad \text{and} \quad \log_y(x^{4y}) = 10$$

We are to find xy .

Step 1: Simplify the first equation

$$\log_x(y^x) = 10$$

Using logarithmic identities:

$$\log_x(y^x) = x \log_x y$$

So:

$$x \log_x y = 10 \quad \text{(1)}$$

Step 2: Simplify the second equation

$$\log_y(x^{4y}) = 10$$

Using the identity $\log_y(a^b) = b \log_y a$:

$$\log_y(x^{4y}) = 4y \log_y x$$

So:

$$4y \log_y x = 10 \quad \text{(2)}$$

Step 3: Let $a = \log_x y$

We know from equation (1):

$$x \log_x y = 10 \Rightarrow xa = 10 \Rightarrow a = \frac{10}{x} \quad \text{(3)}$$

Also, recall that $\log_x y = a$ implies $\log_y x = \frac{1}{a}$ by the reciprocal identity.

From equation (2):

$$4y \cdot \log_y x = 10 \Rightarrow 4y \cdot \frac{1}{a} = 10 \Rightarrow \frac{4y}{a} = 10 \Rightarrow \frac{y}{a} = \frac{10}{4} = \frac{5}{2} \Rightarrow y = \frac{5}{2}a \quad \text{(4)}$$

Step 4: Substitute equation (3) into equation (4)

We have:

$$a = \frac{10}{x} \quad y = \frac{5}{2}a$$

Substitute a :

$$y = \frac{5}{2} \cdot \frac{10}{x} = \frac{50}{2x} = \frac{25}{x}$$

So now:

$$y = \frac{25}{x}$$

Step 5: Find xy

We now have:

$$xy = x \cdot \frac{25}{x} = 25$$

So, $xy = 25$

Now, just to be safe, verify that such x and y exist with $x > 1, y > 1$.

From $y = \frac{25}{x}$, and $x > 1$, then $y > 1$ as long as $x < 25$. But we can find such x .

From earlier:

- $a = \frac{10}{x}$
- $y = \frac{25}{x}$
- $a = \log_x y$

Let's check:

$$\log_x y = \frac{\log y}{\log x}$$

We want to verify this is consistent with $a = \frac{10}{x}$

So:

$$\frac{\log y}{\log x} = \frac{10}{x}$$

But $y = \frac{25}{x}$, so:

$$\frac{\log(25/x)}{\log x} = \frac{\log 25 - \log x}{\log x} = \frac{\log 25}{\log x} - 1$$